INCH-POUND
MIL-DTL-24640/10F
8 November 2011
SUPERSEDING
MIL-C-24640/10E
26 June 2008

DETAIL SPECIFICATION SHEET

CABLE, ELECTRIC, -20 TO +105 °C, 600 VOLTS, LIGHTWEIGHT, MULTI-CONDUCTOR, 16 AWG, TYPES MXSO AND MXSOW

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24640.

Construction, Non-Watertight (MXSO) and Watertight (MXSOW)

First	-	16 AWG, tin-coated copper conductor. Non-watertight constructions shall be constructed using 19 strands. Watertight constructions shall be constructed with 7 strands.
Second	-	Insulation material and basic construction shall be in accordance with SAE-AS81044/12, and shall be tested in accordance with the component wire ratings and requirements listed on this specification sheet. Conformance testing of components shall be in accordance with SAE-AS81044 prior to cabling. Identification in accordance with Method 1, component wire shall be white.
Third	-	The required number of singles shall be cabled together. Cabling sequence shall be consecutive, starting with number 1 from center outward. Fillers may be used, as necessary, to form a round cable.
Fourth	-	Optional binder tape.
Fifth	-	Braided shield, tin-coated copper strands. Braid must be constructed to meet surface transfer impedance requirements as listed herein.
Sixth	-	Optional binder tape.
Seventh	-	Cross-linked polyolefin jacket (see table I for thickness). NOTE: Manufacturer's identification tape or thread is required (see 3.7.3).

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TABLE I. Construction details.

PIN M24640/10	Type number	Number of component wires	Jacket thickness (inch) (min.)	Cable diameter (inch)		Cable weight (pounds per 1000 feet) (max.)
				(min.)	(max.)	
-01UO	MXSO-2	2	0.034	0.282	0.304	76.4
-02UO	MXSO-9	9	0.041	0.424	0.458	180.0
-03UO	MXSO-21	21	0.044	0.552	0.596	329.0
-04UO	MXSO-37	37	0.044	0.644	0.694	497.0
-05UO	MXSOW-2	2	0.034	0.282	0.304	83.0
-06UO	MXSOW-9	9	0.041	0.424	0.458	200.0
-07UO	MXSOW-21	21	0.044	0.552	0.596	365.0
-08UO	MXSOW-37	37	0.044	0.644	0.694	550.0

REQUIREMENTS:

Qualification required.

Construction details: See table I.

Component wire: The component wire shall be tested in accordance with the methods listed in SAE-AS81044 and SAE-AS81044/12 and the following requirements.

Temperature rating	150 °C (302 °F) (max.) conductor temperature
Voltage rating	600 volts (root mean square) at sea level
Finished component diameter:	
Non-watertight	0.072±0.003 inch
Watertight	0.073±0.003 inch
Accelerated aging	Oven temperature, 300±2 °C (572±3.6 °F) for 6 hours
Blocking	150±2 °C (302±3.6 °F)
Flammability	30 seconds (max.)
	3.0 inches (max.)
	No flaming of tissue paper
Humidity resistance	5000 megohms for 1000 feet (min.) insulation resistance after humidity exposure
Dielectric test (100% test)	
Primary insulation	Impulse test - 6.0 kV (peak) or 3 kHz spark test - 1.5kV root mean square
Component wire	Impulse test - 8.0 kV (peak) or 3 kHz spark test - 5.5kV root mean square
Insulation resistance	5000 megohms for 1000 feet (min.)
Life cycle	Oven temperature, 200±2 °C (392±3.6 °F) for 168 hours

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Physical properties of primary insulation	
Tensile strength	2500 pounds per square inch (psi) (min.)
Elongation	150 percent (min.)
Shrinkage	0.125 inch (max.) at 300±2 °C (572±3.6 °F)
Surface resistance	500 megohms inches (min.), initial and final readings
Thermal shock	Oven temperature, 150±2 °C (302±3.6 °F)
Maximum change in measurement	0.060 inch
Wet dielectric test	2500 volts (root mean square)
Wrap test	Mandrel wrap test required

INSPECTION:

Basic Electricals:

Conductor resistance (ohms/1000 feet at 25 °C, max.)	5.15
Voltage withstand (volts, root mean square, min.)	
Conductor to conductor	2000
Conductor to shield	1000
Shield to shield	Not applicable
Insulation resistance (megohms/1000 feet, min.)	2500
Conductor and shield continuity	No failure
Jacket flaws	No failure

Group A:

Visual and dimensional	No failure
Capacitance:	
Mutual capacitance (at 1 megahertz, picofarads/foot, max.)	Not applicable
Capacitance unbalance (percent, max.)	Not applicable
Characteristic impedance (at 1 megahertz, ohms)	Not applicable
Watertightness (Type MXSOW only) (see table IX of MIL-DTL-24640 for limits of water leakage)	No failure
Weight	No failure

Group B:

Cold bending	No failure
Drip (Type MXSOW only)	No failure
Cross-linked proof test (percent, max.)	
Jacket	50

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Tear (lb/in thickness, min.)	35
Physicals (unaged)	
Jacket	
Tensile strength (lb/in^2, min.)	1300
Elongation (percent, min.)	160
Attenuation (at 3 megahertz, decibels/100 feet, max.)	Not applicable
Gas flame (1 hour)	Not applicable

Group C:

Physicals (aged), air oven	
Jacket	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Permanence of printing (conductor insulation, cycles, Method 1 only, min.)	25
Permanence of printing (jacket, cycles, min.)	125
Heat distortion (percent of unaged, max.)	30
Cable sealant removability (Type MXSOW only)	No failure
Shrinkage	No failure
Surface transfer impedance	
Milliohms per meter (max.)	700
EMP response (decibels, min.)	60

Group D:

Flame propagation	No failure

Qualification Inspection:

Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:

Aging and compatibility (cable)	No failure
Acid gas equivalent (percent, max.)	
Jacket	2
Fillers	2
Insulation	18
Halogen content (percent, max.)	
Jacket	0.2
Fillers	0.2

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Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50
Smoke index (max.)	
Jacket	25
Fillers	45
Insulation	10
Toxicity index (max.)	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - Type A (hardness, min.)	80
Weathering (jacket)	No failure
Fungus resistance	No failure
Abrasion resistance	No failure
Gas flame (3 hour)	Not applicable
Component wire	No failure

UNIT ORDERING LENGTHS:

Type and size	Feet (nominal)	
MXSO-2, MXSO-9,	2 & 9	500
MXSO-21, MXSO-37, MXSOW-2, MXSOW-9, MXSOW-21, MXSOW-37	21 & 37	250

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army – MI
Navy – SH

Preparing Activity:
Navy – SH
(Project 6145-2009-078)

Review Activities:
Army – AV, CR, CR4
Navy – CG, EC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.daps.dla.mil.

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